



Model : Power-Xtra PX603048 3.7V 900 mAh Li-Polymer Battery with connector and PCM(1.5A)

Ver: A2

NO: 900.869.503.201

PX603048

Battery Spec

Model: PX-603048

Customer P/N: 900.869.503.201

Cell Type: PX603048

Nominal Voltage: 3.7V

Capacity: 900mAh

Draft	Checking	Approved	Customer Confirmation
Dora	Peter		



Model : Power-Xtra PX603048 3.7V 900 mAh Li-Polymer Battery with connector and PCM(1.5A)

Ver: A2

NO: 900.869.503.201

1. Revision History

Revision	Date	Editor	Contents
A0	2018-06-14	Dora	Draft
A1	2018-06-19	Dora	增加 NTC 参数、条码内容
A2	2018-06-21	Dora	更改出线长度、插头方向

2. Product Specification

(Single cell)

No.	Item	General Parameter		Remark
1	Rated Capacity	Typical	900mAh	Standard discharge (0.2C) after Standard charge
		Minimum	880mAh	
2	Nominal Voltage	3.7V		Mean Operation Voltage
3	Voltage at end of Discharge	2.75V		Discharge Cut-off Voltage
4	Charging Voltage	4.2±0.03V		
5	Internal Impedance	≤200mΩ		Internal resistance measured at AC 1KHZ after 50% charge The measure must uses the new batteries that within one week after shipment and cycles less than 5 times
6	Weight	About 23 g		
7	Standard charge	Constant Current 0.2C Constant Voltage 4.2V 0.01 C cut-off		
8	Standard discharge	Constant current 0.2C end voltage 2.75V		
9	Fast charge	Constant Current 1.0C Constant Voltage 4.2V 0.01C cut-off		
10	Fast discharge	Constant current 1.0C end voltage 2.75V		
11	Maximum Continuous Charge Current	1.0C		
12	Maximum Continuous Discharge Current	1.0C		
13	Operation Temperature Range	Charge : 0~45°C		60±25%R.H.
		Discharge: -20~60°C		Bare Cell
14	Storage Temperature Range	Less than 1 year: -20~25°C		60±25%R.H.
		less than 3 months: -20~40°C		at the shipment state
15	Single cell	Length (L)	48.0±0.5mm	Initial Dimension
		Width (W)	30.0±0.5mm	
		Thickness (T)	6.0±0.2mm	

3. Performance And Test Conditions

3.1 Standard Test Conditions

Test should be conducted with new batteries within one week after shipment from our factory and the cells shall not be cycled more than five times before the test. Unless otherwise specified, test and measurement shall be done under temperature of $20\pm5^{\circ}\text{C}$ and relative humidity of 45~85%. If it is judged that the test results are not affected by such conditions, the tests may be conducted at temperature $15\sim30^{\circ}\text{C}$ and humidity 25~85%RH.

3.2 Measuring Instrument or Apparatus

3.2.1 Dimension Measuring Instrument

The dimension measurement shall be implemented by instruments with equal or more precision scale of 0.01mm.

3.2.2 Voltmeter

Standard class specified in the national standard or more sensitive class having inner impedance more than $10\text{k}\Omega/\text{V}$

3.2.3 Ammeter

Standard class specified in the national standard or more sensitive class. Total external resistance including ammeter and wire is less than 0.01Ω .

3.2.4 Impedance Meter

Impedance shall be measured by a sinusoidal alternating current method(1kHz LCR meter).

3.3 Appearance

There shall be no such defect as flaw, crack, rust, leakage, which may adversely affect commercial value of battery.

3.4 Temperature Dependence of discharge capacity

Table 3 (3)

Discharge Temperature	-10°C	0°C	23°C	60°C
Discharge Capacity (0.2C)	50%	80%	100%	95%

3.5 Cycle Life and Leakage-Proof

Table 4

No.	Item	Criteria	Test Conditions
1	Cycle Life (0.5C)	Higher than 70% of the Initial Capacities of the Cells	Carry out 500cycle Charging/Discharging in the below condition. ◆Charge: Standard Charge ◆Discharge:0.5C to 2.75 V ◆Rest Time between charge/discharge:30min. ◆Temperature:20±5°C
2	Leakage-Proof	No leakage (visual inspection)	After full charge with standard charge, store at $55\pm3^{\circ}\text{C}$, 60±10%RH for 1 week.

4. Mechanical characteristics and Safety Test for Cell

Table 5

(Mechanical characteristics)

No.	Items	Test Method and Condition	Criteria
1	Vibration Test	After standard charging, fixed the cell to vibration table and subjected to vibration cycling that the frequency is to be varied at the rate of 1Hz per minute between 10Hz an 55Hz, the excursion of the vibration is 1.6mm. The cell shall be vibrated for 30 minutes per axis of XYZ axes.	No leakage No fire
2	Drop Test	The cell is to be dropped from a height of 1 meter twice onto concrete ground.	No explosion, No fire, no leakage.
Item	Battery Condition	Test Method	Requirements
Crush	Fresh, Fully charged	Crush between two flat plates. Applied force is about 13kN(1.72Mpa) for 30min.	No explosion, No fire
Short Circuit	Fresh, Fully charged	Each test sample battery, in turn, is to be short-circuited by connecting the (+) and (-) terminals of the battery with a Cu wire having a maximum resistance load of 0.1Ω. Tests are to be conducted at room temperature(20±2°C).	No explosion, No fire The Temperature of the surface of the Cells are lower than 150°C
Short Circuit	Fresh, Fully charged	Each test sample battery, in turn, is to be short-circuited by connecting the (+) and (-) terminals of the battery with a Cu wire having a maximum resistance load of 0.1Ω. Tests are to be conducted at temperature(60±2°C).	No explosion, No fire The Temperature of the surface of the Cells are lower than 150°C
Impact	Fresh, Fully charged	A 56mm diameter bar is inlayed into the bottom of a 10kg weight. And the weight is to be dropped from a height of 1m onto a sample battery and then the bar will be across the center of the sample.	No explosion, No fire
Forced Discharge	Fresh, Fully charged	Discharge at a current of 1.0C for 2.5h.	No explosion, No fire
Nail Pricking (3mm)	Fresh, Fully charged	Prick through the sample battery with a nail having a diameter of 3mm and remain 2h.	No explosion, No fire

5. Protection circuit

(PCM Standard)

Item	Symbol	Content	Criterion
Current	IDP	Max. Charging Current	1.5A
		Max. Discharging Current	1.5A
Over charge Protection	VDET1	Over charge detection voltage	4.30±0.05V
	tVDET1	Over charge detection delay time	50–270ms
	VREL1	Over charge release voltage	4.15±0.05V
Over discharge protection	VDET1	Over discharge detection voltage	3.0±0.05V
	tVDET1	Over discharge detection delay time	5-25ms
	VREL1	Over discharge release voltage	3.10±0.1V
Over current protection	VDET3	Over current detection voltage	150±30mV
	IDP	Over current detection current	3.5±1.5A
	tVDET3	Detection delay time	5-26ms
		Release condition	Cut load
Short protection		Detection condition	Exterior short circuit
	TSHORT	Detection delay time	5-50ms
		Release condition	Cut short circuit
Interior resistance	RDS	Main loop electrify resistance	VC=4.2V,RDS≤70mΩ
Current consumption	IDD	Current consume in normal operation	3.0μA Type 6.0μA Max

6. Handling of Cells

6.1 Consideration of strength of film package

1) Soft Aluminium foil

Easily damaged by sharp edge parts such as pins and needles, Ni-tabs, comparing with metal-can-cased LIB.

2). Sealed edge may be damaged by heat above 100°C, bend or fold sealed edge.

6.2 Prohibition short circuit

Never make short circuit cell. It generates very high current which causes heating of the cells and may cause electrolyte leakage, gassing or explosion that are very dangerous.

The Power-Xtra tabs may be easily short-circuited by putting them on conductive surface. Such outer short circuit may lead to heat generation and damage of the cell.

An appropriate circuitry with PCM shall be employed to protect accidental short circuit of the battery pack.



6.3.Mechanical shock

Power-Xtra cells have less mechanical endurance than metal-can-cased LIB.

Falling, hitting, bending, etc. may cause degradation of Power-Xtra characteristics.

6.4 Handling of tabs

The battery tabs are not so stubborn especially for aluminum tab.

Don't bend tab.

Do not bend tabs unnecessarily.

7. Storing the Batteries

The batteries should be stored at room temperature, charged to about 30% to 50% of capacity.

We recommend that batteries be charged about once per half a year to prevent over discharge.

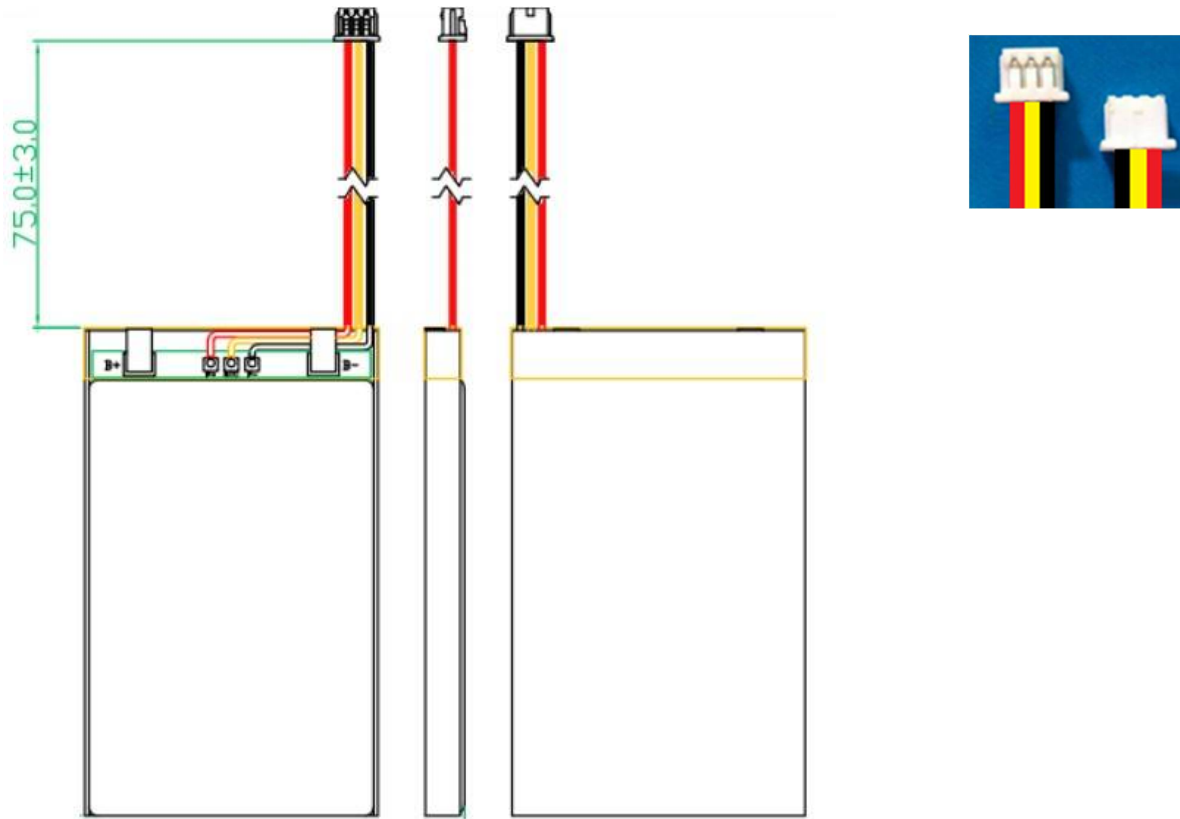
POWER-XTRA

Model : Power-Xtra PX603048 3.7V 900 mAh Li-Polymer Battery with connector and PCM(1.5A)

Ver: A2

NO: 900.869.503.201

8. Dimension



Dimensions (Units: mm)	PCM	Normal PCM (1.5A)
	Length Cable (L)	75±5mm
	Height (H)	50.0±1mm
	Width (W)	30.5±1mm
	Thickness (T)	6.2±0.5mm
	Cable	UL1007#26AWG
	Plug	Molex 51021-0300 NTC,
	NTC	10K

POWER-XTRA

Model : Power-Xtra PX603048 3.7V 900 mAh Li-Polymer Battery with connector and PCM(1.5A)

Ver: A2

NO: 900.869.503.201

9. Drawing of Label

PET, 2D(Data Matrix): "8680187004729"。YY, MM, 如: 1604 (2016 年 07 月)。



10. Drawing Packing

整齐装托盘，内置防潮袋，每箱不超 10KG；客户定制 Logo 纸箱，外箱 Logo 格式如下：

POWER-XTRA

ENA-13 Bar code :

130*100mm:

← PO NO.	Order 16-8
← MODEL NO.	900.869.503.201
← QTY	500PCS
DATE	YYYY-MM-DD
Made in China	
 8 680187 004729	

根据每次订单更改

根据每箱数量更改

根据出货日期更改